

FOD FPGA Project — HLS Synthesis Summary

Baseline synthesis of the quantized MLP accelerator targeting the PYNQ-Z2 / Zynq-7020.

1. Synthesis Status

The Vitis HLS synthesis completed successfully. The synthesized design corresponds to the integer MLP implementation used in the preceding C simulation, where all 6 test vectors passed.

2. Target Device

Parameter	Value
FPGA device	xc7z020-clg400-1
Target platform	PYNQ-Z2 / Zynq-7020
Target clock period	10.00 ns
Target frequency	100 MHz

3. Performance Estimates

Metric	Result
Estimated clock period	6.948 ns
Target clock period	10.00 ns
Approx. frequency corresponding to 6.948 ns	≈ 144 MHz
Total latency	274 clock cycles
Absolute latency @ 100 MHz	2.740 μs

Latency Breakdown

Block	Cycles	Time @ 100 MHz
Layer 1 — Dense 64→32	116	1.16 μs
Layer 2 — Dense 32→16	100	1.00 μs
Layer 3 — Dense 16→4	11	0.11 μs
Argmax	5	0.05 μs
Total	274	2.74 μs

4. FPGA Resource Utilization

Resource	Used	Available	Utilization
BRAM	4	280	≈ 1%
DSP	70	220	≈ 32%
FF	7,214	106,400	≈ 6%
LUT	8,728	53,200	≈ 16%
URAM	0	0	—

5. DSP Usage Breakdown

Component	DSPs
Layer 1	33

Layer 2	17
Layer 3	9
Other generated arithmetic	11
Total	70

6. Baseline Architecture

64 INT8 features → Dense 64→32 → ReLU + requantization → Dense 32→16 → ReLU + requantization → Dense 16→4 → INT32 logits → Argmax

7. Project Status After Synthesis

Stage	Status
RGB feature extraction / Stage 1	Complete
INT8 quantization	Complete
Python integer model	Verified — 98.30% test accuracy
HLS C++ implementation	Complete
HLS C simulation	Complete — 6/6 PASS
HLS synthesis	Complete — SUCCESS

Next planned stage: package the synthesized MLP as an HLS IP and integrate it into the PYNQ-Z2 Vivado design, including the interface for supplying the 64 INT8 features.

Note: This summary is based on the uploaded Vitis HLS synthesis report and the synthesis results discussed during the project workflow.